

Electromagnetic junction conditions for non-null hypersurfaces

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Abstract

This paper derives the electromagnetic junction conditions in general relativity when two spacetimes are joined across a non-null hypersurface of constant signature. In addition, these junction conditions, along with Israel junction conditions, are applied to study an idealized star with surface charge density.

1 Introduction

In general relativity, junction conditions are boundary conditions at spacetime boundaries that arise from joining two distinct spacetimes. These spacetime boundaries are described by a hypersurface Σ that, in the most general cases, has different signatures at different points [1]. The most studied junction conditions are those of the metric and its derivatives, as the metric defines the spacetime geometry. For a simple introduction to the metric junction conditions in general relativity, see [2].

While the metric has rightfully been the primary focus of studies on junction conditions in general relativity, there are other areas of research. Examples include studies of scalar-tensor theories in modified gravity where the scalar field junction conditions must be considered alongside the metric junction conditions [3], and gravitoelectromagnetic junction conditions [4]. The metric junction conditions are also used in brane world cosmology, where our universe is a hypersurface (brane) in a higher dimensional spacetime (bulk) [5]. In these theories, gravity in the bulk can induce fields such as an electromagnetic field in the brane [6].

In this paper, we consider electromagnetic junction conditions in the context of general relativity. We derive these junction conditions using distributional methods, which are commonly used in modern derivations of the Israel junction conditions [7]. Like the Israel junction conditions, our electromagnetic junction conditions will also be limited to hypersurfaces that are either spacelike or timelike.

There are few relevant papers in the literature that explore electromagnetic junction conditions in general relativity. The most relevant one is by Rawson-Harris [8], where the covariant electromagnetic boundary conditions (junction conditions) are derived for non-null hypersurfaces. While our approach is different, we obtain similar results. Our results

are also simpler because we further reduce the junction conditions to a form that primarily depends on the intrinsic coordinates of a hypersurface.

In Section 2 we briefly introduce the Israel junction conditions. The electromagnetic junction conditions are derived in Section 3 using the Maxwell equations in curved spacetime. In Section 4 we calculate both the metric and the electromagnetic junction conditions for a static spherically symmetric star of constant density with a surface charge density such that the exterior spacetime will be Reissner-Nordström. We conclude the paper in Section 5 with a discussion of our results.

2 Geometry

In this paper, we work in (3+1)-dimensional spacetime with the metric signature $(-, +, +, +)$. Let $\mathcal{V}$ be a spacetime region containing two distinct spacetime regions $\mathcal{V}^+$ and $\mathcal{V}^-$ separated by a hypersurface Σ . In $\mathcal{V}^\pm$ we have distinct coordinates $x^{\alpha\pm}$, but we assume that we can use coordinate x^α , that is distinct from $x^{\alpha\pm}$, such that it overlaps with $x^{\alpha\pm}$ in open region around Σ . We also assume that Σ is not null, since we will not be investigating the case for null Σ .

A hypersurface Σ can be described by a function

$$\Phi(x^\alpha) = 0, \quad (1)$$

or by a parametric equation

$$x^\alpha = x^\alpha(y^a), \quad (2)$$

with y^a being intrinsic coordinates of Σ [7]. (Throughout this paper, we use Latin indices to refer to the 3D coordinates of Σ , and Greek indices to refer to the 4D coordinates in $\mathcal{V}$.) Σ is characterized by its holonomic basis vectors, the normal vector, the induced metric, and the extrinsic curvature. The holonomic basis vectors are

$$e_a^\alpha = \frac{\partial x^\alpha}{\partial y^a}. \quad (3)$$

The normal vector is defined as [7]

$$n^\alpha = \frac{\varepsilon \Phi_{,\alpha}}{|g^{\mu\nu} \Phi_{,\mu} \Phi_{,\nu}|}, \quad (4)$$

where

$$n^\alpha n_\alpha = \varepsilon \equiv \begin{cases} -1 & \text{if } \Sigma \text{ is spacelike,} \\ +1 & \text{if } \Sigma \text{ is timelike.} \end{cases} \quad (5)$$

The normal vector is defined such that it points towards the direction of increasing Φ such that $n^\alpha \Phi_{,\alpha} > 0$. Usually we can figure n^α from the conditions $n^\alpha n_\alpha = \varepsilon$ and $n^\alpha u_\alpha = 0$, where u^α is four-velocity of an observer co-moving with Σ . For stationary Σ , $n^\alpha u_\alpha = 0$ is trivial, in which case $n^\alpha n_\alpha = \varepsilon$ is enough to determine n^α . The induced metric of Σ is

$$h_{ab} = g_{\alpha\beta} e_a^\alpha e_b^\beta, \quad (6)$$

and the extrinsic curvature is

$$K_{ab} = n_{\alpha;\beta} e_a^\alpha e_b^\beta. \quad (7)$$

The extrinsic curvature can also be defined in terms of the Lie derivative of the metric along the normal n :

$$K_{ab} = \frac{1}{2}(\mathcal{L}_n g_{\alpha\beta}) e_a^\alpha e_b^\beta. \quad (8)$$

The induced metric and the extrinsic curvature are important for formulating the junction conditions for the metric field in a coordinate-independent way.

2.1 Israel junction conditions

Israel junction conditions are the junction conditions for the metric field when the boundary between two spacetimes is a non-null regular hypersurface Σ . The first and the second junction conditions are [7, 9]

$$[h_{ab}] = 0, \quad [K_{ab}] = 0. \quad (9)$$

The square brackets denote the difference in quantities across Σ :

$$[A] \equiv A^+|_\Sigma - A^-|_\Sigma, \quad (10)$$

the labels for spacetime regions are such that n^α points towards $\mathcal{V}^+$.

The second junction condition $[K_{ab}] = 0$ is violated if there is a singular (thin) layer of matter at Σ . This leads to Σ having a stress tensor according to Lanczos equation:

$$S_{ab} = -\frac{\varepsilon}{8\pi}([K_{ab}] - [K]h_{ab}), \quad (11)$$

where $K = h^{ab}K_{ab}$ [2, 7]. This model with the second junction condition violated is usually called the thin shell formalism.

Now that we have a basic understanding of the geometry of hypersurfaces and the Israel junction conditions, we are ready to derive the electromagnetic junction conditions.

3 Electromagnetic junction conditions

In this section we work with the geometry of spacetime described in the previous section. We have a hypersurface Σ separating two spacetimes regions $\mathcal{V}^+$ and $\mathcal{V}^-$, where the Maxwell's equations apply. Maxwell's equations in curved spacetimes are

$$F^{\alpha\beta}{}_{;\beta} = 4\pi J^\alpha, \quad (12)$$

$$F_{[\alpha\beta;\gamma]} \equiv F_{\alpha\beta;\gamma} + F_{\beta\gamma;\alpha} + F_{\gamma\alpha;\beta} = 0, \quad (13)$$

where $F^{\alpha\beta}$ is the electromagnetic tensor defined via the vector potential A^μ [10]:

$$F_{\alpha\beta} = A_{\beta;\alpha} - A_{\alpha;\beta}. \quad (14)$$

We are using the convention of Misner, Thorne, and Wheeler where the electric and magnetic fields are defined by

$$F^{0j} = -F^{j0} = E^j, \quad F^{jk} = \epsilon^{jkl} B^l, \quad (15)$$

where ϵ^{jkl} is the Levi-Civita symbol [10].

We formulate the electromagnetic junction conditions by investigating the necessary conditions for obtaining valid distributional solutions to the Maxwell's equations. For the solutions to be valid they also need to produce a distributionally valid energy-momentum tensor for electromagnetism, which is defined by [10]

$$T^{(EM)}_{\alpha\beta} = \frac{1}{4\pi} \left(F_{\alpha\mu} F_{\beta}^{\mu} - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} g_{\alpha\beta} \right). \quad (16)$$

3.1 First electromagnetic junction conditions

We start with a distributional vector potential as the solution to the Maxwell's equations

$$A_{\mu} = \Theta(l) A_{\mu}^{+} + \Theta(-l) A_{\mu}^{-}. \quad (17)$$

Using the definition for the electromagnetic tensor (14) we get

$$F_{\alpha\beta} = \Theta(l) F_{\alpha\beta}^{+} + \Theta(-l) F_{\alpha\beta}^{-} + \varepsilon \delta(l) ([A_{\beta}] n_{\alpha} - [A_{\alpha}] n_{\beta}). \quad (18)$$

This will produce a term proportional to $\Theta(l)\delta(l)$ in the energy-momentum tensor, as it is quadratic in $F_{\alpha\beta}$. To avoid this distributionally invalid term, we require the constraint $[A_{\beta}] n_{\alpha} - [A_{\alpha}] n_{\beta} = 0$. This constraint leads directly to our first electromagnetic junction condition:

$$[A_a] = 0. \quad (19)$$

This means that the vector potential must have continuous tangential components while the normal component may have jump across Σ . The direct consequence of this is that the junction condition is automatically expressed in terms of the intrinsic coordinates of Σ .

With the first electromagnetic junction condition in place, the distributional electromagnetic tensor simplifies to

$$F_{\alpha\beta} = \Theta(l) F_{\alpha\beta}^{+} + \Theta(-l) F_{\alpha\beta}^{-}. \quad (20)$$

3.2 Second electromagnetic junction condition

The covariant derivative of the electromagnetic tensor is

$$F^{\alpha\beta}_{;\beta} = \Theta(l) F^{+\alpha\beta}_{;\beta} + \Theta(-l) F^{-\alpha\beta}_{;\beta} + \varepsilon \delta(l) [F^{\alpha\beta}] n_{\beta}. \quad (21)$$

The Maxwell's equation (12) implies

$$J^{\alpha} = \Theta(l) J^{+\alpha} + \Theta(-l) J^{-\alpha} + \delta(l) I^{\alpha}, \quad (22)$$

where $I^\alpha = J_\Sigma^\alpha$ is the four-current density on Σ . This means that the surface current density on Σ is given by

$$\varepsilon[F^{\alpha\beta}]n_\beta = 4\pi I^\alpha \quad . \quad (23)$$

This is our second electromagnetic junction condition, and we can simplify it in terms of intrinsic coordinates of Σ by using the antisymmetric properties of $F^{\alpha\beta}$.

By contracting (23) with the normal vector n_α and using antisymmetric properties of the electromagnetic tensor we get

$$[F^{\beta\alpha}]n_\alpha n_\beta = -[F^{\beta\alpha}]n_\alpha n_\beta \quad . \quad (24)$$

This is only true for $[F^{\alpha\beta}]n_\alpha n_\beta = 0$, meaning surface current doesn't have any component along the normal direction. This reduces the second electromagnetic junction condition to

$$\varepsilon[F^a] = 4\pi I^a \quad , \quad (25)$$

where $F^a = F^{a\beta}n_\beta$.

3.3 Third electromagnetic junction condition

Plugging the electromagnetic tensor (20) into the Maxwell's equation (13) gives us

$$0 = [F_{\alpha\beta}]n_\gamma + [F_{\beta\gamma}]n_\alpha + [F_{\gamma\alpha}]n_\beta \quad , \quad (26)$$

the third and final electromagnetic junction condition. This equation is trivial when two indices are the same, due to antisymmetry. This simplifies the third junction condition to

$$[F_{ab}] = 0 \quad . \quad (27)$$

The simplest way to see this is to work in a basis with holonomic basis vectors of Σ and n^α as the basis vectors, where this becomes trivial.

3.4 Energy-momentum tensor

As a consequence of the first electromagnetic junction conditions, the electromagnetic energy-momentum tensor has the form

$$T^{(EM)}_{\alpha\beta} = \Theta(l)T^+_{\alpha\beta} + \Theta(-l)T^-_{\alpha\beta} \quad , \quad (28)$$

where $T^\pm_{\alpha\beta} = \frac{1}{4\pi} \left(F^\pm_{\alpha\mu} F^\pm_{\beta}{}^{\mu} - \frac{1}{4} F^\pm_{\mu\nu} F^\pm{}^{\mu\nu} g^\pm_{\alpha\beta} \right)$ with the $\pm$ signs describing the quantities in different sides Σ . This result tells us that the energy-momentum tensor produced by an electromagnetic field cannot be singular even if there is charge density at Σ and the electromagnetic tensor $F_{\alpha\beta}$ is discontinuous. This is not surprising since $T^{\alpha\beta}$ is quadratic in $F_{\alpha\beta}$ —this restricts $F_{\alpha\beta}$ to be regular at Σ —but it is an interesting contrast to the energy-momentum tensor of matter, which can be singular at Σ .

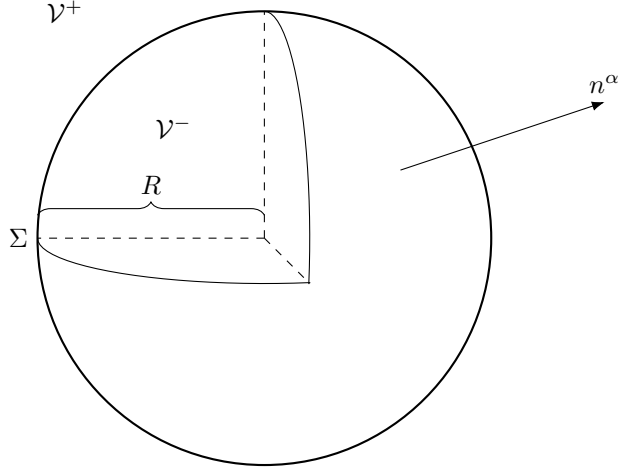


Figure 1: Visual representation of the star: manifolds $\mathcal{V}^-$ and $\mathcal{V}^+$ and the boundary Σ between them. The vector n^α is the normal vector that points to the spacetime $\mathcal{V}^+$ from the hypersurface Σ .

4 A star with a surface charge density

As an example, we present a calculation of junction conditions for a star that has a uniform charge density at the surface while the rest of the star is electrically neutral.

Let us consider the following situation, as depicted in figure 1. We have an exterior Reissner-Nordström (RN) spacetime $\mathcal{V}^+$, joined to an interior spacetime $\mathcal{V}^-$, representing a star with constant density ρ_* and total charge Q , which is entirely confined to its surface. The hypersurface Σ separating these regions is a time-like hypersurface. The radial coordinate r is an affine parameter that is smooth across the boundary and the same on both sides. The coordinates on Σ can be straightforwardly chosen to be $y^a = (t, \theta, \phi)$.

We assume an idealized star that is static with constant density ρ_* and is spherically symmetric. The following metric in Schwarzschild coordinates describes the inner part of this star

$$ds^2 = - \left[\frac{3}{2} \left(1 - \frac{2M_*}{R} \right)^{\frac{1}{2}} - \frac{1}{2} \left(1 - \frac{2M_* r^2}{R^3} \right)^{\frac{1}{2}} \right]^2 dt^2 + \left(1 - \frac{2m(r)}{r} \right)^{-1} dr^2 + r^2 d\Omega^2, \quad (29)$$

where $m(r) = \frac{4\pi}{3} \rho_* r^3$ is the mass enclosed within radius r , R is the radius of the star, and $M_* = m(R)$ is the total mass contributed by the density of the star [10]. Outside the star, in $\mathcal{V}^+$, the metric is RN

$$ds^2 = -f_{RN} dt^2 + f_{RN}^{-1} dr^2 + r^2 d\Omega^2, \quad (30)$$

with $f_{RN} = 1 - \frac{2M}{r} + \frac{Q^2}{r^2}$, where M is the total gravitational mass distinct from M_* due to the contribution of the total charge Q to the spacetime curvature.

Let us first calculate the Israel junction conditions to see the conditions for joining the spacetimes. We must first compute the induced metric and the extrinsic curvature to

calculate the Israel junction conditions. The induced metric is given by

$$h_{ab} = g_{\alpha\beta} e_a^\alpha e_b^\beta, \quad (31)$$

which means that the metrics on the two sides of Σ (at $r = R$) are:

$$ds^2|_{\Sigma^-} = - \left(1 - \frac{2M_\star}{R}\right) dt^2 + R^2 d\Omega^2, \quad (32)$$

$$ds^2|_{\Sigma^+} = -f_{RN} dt^2 + R^2 d\Omega^2. \quad (33)$$

The first Israel junction condition $[h_{ab}] = 0$ gives us

$$M = M_\star + \frac{1}{2} \frac{Q^2}{R}. \quad (34)$$

This relation shows that the total mass has a charge contribution in addition to the stellar mass $M_\star = \frac{4\pi}{3} \rho_\star R^3$, and this contribution is singular just like the charge distribution itself.

Using equation (8) we calculate the extrinsic curvature to be

$$K_{tt}^- = -\frac{M_\star}{R^2 \sqrt{g_{RR}^-}} \quad K_{tt}^+ = \frac{1}{R^2 \sqrt{g_{RR}^+}} \left(-M + \frac{Q^2}{R}\right) \quad (35)$$

$$K_{\theta\theta}^- = \frac{R}{\sqrt{g_{RR}^-}} \quad K_{\theta\theta}^+ = \frac{R}{\sqrt{g_{RR}^+}} \quad (36)$$

$$K_{\phi\phi}^- = \frac{R \sin^2 \theta}{\sqrt{g_{RR}^-}} \quad K_{\phi\phi}^+ = \frac{R \sin^2 \theta}{\sqrt{g_{RR}^+}}, \quad (37)$$

where $g_{RR}^\pm \equiv g_{rr}^\pm|_{r=R}$. We see that the only discontinuous component is K_{tt} , which using (34) simplifies to

$$[K_{tt}] = \frac{1}{2R^2 \sqrt{g_{RR}}} \left(\frac{Q^2}{R}\right), \quad (38)$$

where $\sqrt{g_{RR}} = \sqrt{g_{RR}^\pm}$. Thus the second junction condition is violated and there is a thin shell at Σ .

To see what contributes to this thin shell, let us calculate the energy-momentum tensor of Σ . Using the Lanczos equation (11) and (38) we get the components of S_{ab} :

$$S_{tt} = 0 \quad (39)$$

$$S_{\theta\theta} = -\frac{1}{16\pi} \sqrt{g_{RR}} \left(\frac{Q^2}{R}\right) \quad (40)$$

$$S_{\phi\phi} = -\frac{1}{16\pi} \sqrt{g_{RR}} \left(\frac{Q^2}{R}\right) \sin^2 \theta. \quad (41)$$

The singular charge density produces a singular energy-momentum tensor at Σ , not because of the mass of the charge carriers, which is included in the mass $M_\star$, but from the

charge's contribution to the total mass of the system via equation (34). This equation tells that there is an extra layer effective of mass $M_\Sigma = \frac{1}{2} \frac{Q^2}{R}$ at Σ . This is also evident from the Q^2 dependence of the singular part of the energy-momentum tensor. It is an interesting result that while the electromagnetic fields themselves cannot contribute to the singular part of the energy-momentum tensor, the charge density does so via its effect on the spacetime curvature.

Let us now use the previously derived electromagnetic junction conditions. In $\mathcal{V}^+$ we have the stress tensor $F^{\alpha\beta}$ with the non-zero components $F^{tr} = \frac{Q}{r^2} = -F^{rt}$. In $\mathcal{V}^-$, where there is no electromagnetic field we have $F^{\alpha\beta} = 0$. The first electromagnetic junction condition is trivial because we can set the potential in V^- to be constant such that $A_\alpha^-(r) = A_\alpha^+(R)$. The third electromagnetic junction condition is also trivial since the only non-zero components of the electromagnetic tensor in $\mathcal{V}^+$ are $F^{tr} = -F^{rt}$ and r is the index along the normal, which means $F_{ab}^+ = 0 \Rightarrow [F_{ab}] = 0$. The second electromagnetic junction condition is not trivial since we have a star with a surface charge density:

$$\varepsilon[F^{\alpha\beta}]n_\beta = 4\pi I^\alpha \quad (42)$$

$$\Rightarrow \frac{Q}{R^2} \sqrt{g_{RR}} = 4\pi I^t. \quad (43)$$

This means that the charge density observed by a stationary observer, whose four-velocity is $u_\alpha = \sqrt{-g_{tt}}\delta_{t\alpha}$, is

$$I^t u_t = \frac{Q}{4\pi R^2} \sqrt{g_{RR}} \sqrt{-g_{tt}} = \frac{Q}{4\pi R^2}. \quad (44)$$

This is exactly the surface charge density we had assigned the surface. Thus, a star with a surface charge density satisfies the electromagnetic junction conditions.

5 Discussion

We have derived electromagnetic junction conditions for timelike and spacelike hypersurfaces using distributional methods. These junction conditions only depend on the intrinsic coordinates of the hypersurface. Our coordinate-independent formulation is new compared to the derivation of electromagnetic junction conditions by Rawson-Harris [8]. The advantage of Rawson-Harris's approach is that it works with hypersurfaces that can arbitrarily change signature between timelike and spacelike. The advantage of our approach, in addition to simpler equations, is that it does not require the electromagnetic sources to be independent of time and works for dynamic situations.

This paper doesn't consider null boundaries. I do have calculation for some junction conditions, but have not had time to dig deeper and add them to this paper. As for other limitations, we have not compared our calculations on the star with surface charge density to any other work in the literature. Similarly, the comparisons with other literature on similar topics might be lacking.

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